```
!nvidia-smi
```

```
Tue Aug 12 15:03:13 2025
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 560.35.03              Driver Version: 560.35.03      CUDA Version: 12.6     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   42C    P8              9W /   70W |       1MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   1  Tesla T4                       Off |   00000000:00:05.0 Off |                    0 |
| N/A   40C    P8              9W /   70W |       1MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI        PID   Type   Process name                              GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

Import Libraries

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

Connect Google Drive

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

Load CSV File

```
data = pd.read_csv("/content/drive/MyDrive/Research Projects /Data Science and ML /Code/WQI Calculation Code/New
data.head()
```

	WaterbodyName	Years	SampleDate	Alkalinity-total (as CaCO3)	Ammonia-Total (as N)	BOD - 5 days (Total)	Chloride	Conductivity @25°C	Dissolved Oxygen	ortho Phosphat (as P) unspecifie
0	ABBEYTOWN_010	2023	Feb	314.0	0.033	1.2	27.3	711.0	52.50	0.01
1	Allua	2007	Aug	14.0	0.033	1.2	10.0	71.0	61.85	0.01
2	Allua	2007	Aug	17.0	0.033	1.2	11.5	79.0	61.80	0.01
3	Allua	2007	Aug	18.0	0.033	1.2	11.3	78.0	62.40	0.01
4	Allua	2007	Sep	19.0	0.033	1.2	10.5	76.0	66.05	0.01

5 rows × 22 columns

Null Value Check: Though it was confirmed in Data Wragling Phase

```
null_values = data.isnull().sum()
null_values
```

	0
WaterbodyName	0
Years	0
SampleDate	0
Alkalinity-total (as CaCO3)	0
Ammonia-Total (as N)	0
BOD - 5 days (Total)	0
Chloride	0
Conductivity @25°C	0
Dissolved Oxygen	0
ortho-Phosphate (as P) - unspecified	0
pH	0
Temperature	0
Total Hardness (as CaCO3)	0
True Colour	0
IndWQI	0
IndWQI_Score	0
IWQI	0
IWQI_Score	0
DWQI	0
DWQI_Score	0
AWQI	0
AWQI_Score	0

dtype: int64

```
data = data.dropna()
```

```
null_values = data.isnull().sum()
null_values
```

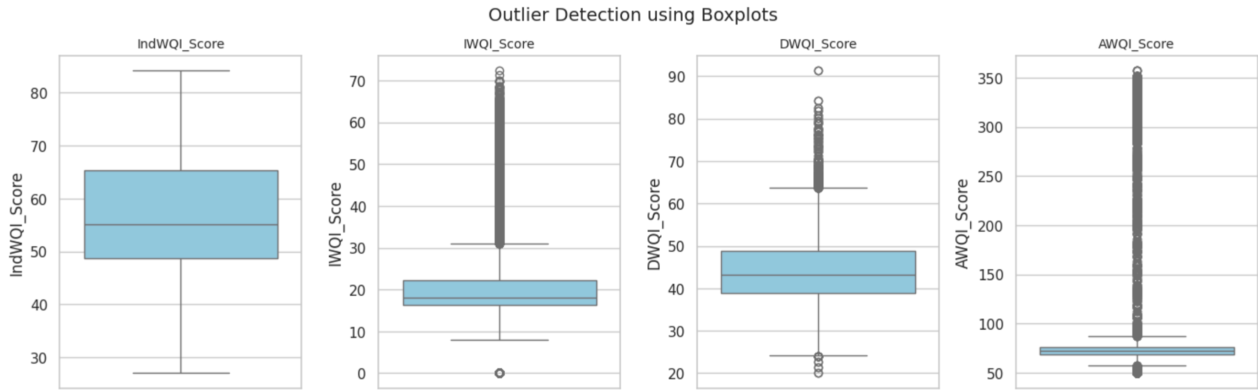
```
DO                0
Ammonia           0
Hardness          0
Temp              0
Alkalinity        0
Conductivity      0
BOD               0
Color             0
orthophosphate    0
pH                0
CCME_WQI          0
CCME_WQI_Score    0
WAWQI_Values      0
WAWQI_Quality     0
IndWQI            0
IndWQI_Score      0
IWQI              0
IWQI_Score        0
DWQI              0
DWQI_Score        0
AWQI              0
AWQI_Score        0
RWQI              0
RWQI_Score        0
dtype: int64
```

```
data.shape
```

```
(29159, 22)
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# List of columns to check for outliers
columns = ['IndWQI_Score',            'IWQI_Score', 'DWQI_Score', 'AWQI_Score']
```

```
# Set plot style
sns.set(style="whitegrid")
plt.figure(figsize=(14, 8))

# Create boxplots for each column
for i, col in enumerate(columns, 1):
    plt.subplot(2, 4, i)  # Adjust the layout: 2 rows, 4 columns
    sns.boxplot(y=data[col], color="skyblue")
    plt.title(col, fontsize=10)
    plt.tight_layout()

plt.suptitle("Outlier Detection using Boxplots", fontsize=14, y=1.02)
plt.show()
```




```
# List of columns to check for outliers
columns = ['CCME_WQI_Score', 'WAWQI_Values', 'IndWQI_Score',
           'IWQI_Score', 'DWQI_Score', 'AWQI_Score', 'RWQI_Score']

# Make a copy to preserve original
data_cleaned = data.copy()

# Loop through each column to remove outliers
for col in columns:
    Q1 = data_cleaned[col].quantile(0.25)
    Q3 = data_cleaned[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Filter out the outliers
    data_cleaned = data_cleaned[(data_cleaned[col] >= lower_bound) & (data_cleaned[col] <= upper_bound)]

# Check the shape before and after
print(f"Original data shape: {data.shape}")
print(f"Cleaned data shape: {data_cleaned.shape}")
```

```
data = data.rename(columns={
    'Dissolved Oxygen': 'DO',
    'Ammonia-Total (as N)': 'Ammonia',
    'Total Hardness (as CaCO3)': 'Hardness',
    'Temperature': 'Temp',
    'Alkalinity-total (as CaCO3)': 'Alkalinity',
    'Conductivity @25°C': 'Conductivity',
    'BOD - 5 days (Total)': 'BOD',
    'True Colour': 'Color',
    'ortho-Phosphate (as P) - unspecified': 'orthophosphate',
    'pH': 'pH'  # stays the same but included for completeness
})
```

```
# Define the input features and output variables
input_features = [
    'Alkalinity', 'Ammonia', 'BOD',
       'Chloride', 'Conductivity', 'DO', 'orthophosphate', 'pH', 'Temp',
       'Hardness', 'Color' ]
```

```
Reg_output = ['IndWQI_Score','IWQI_Score','DWQI_Score', 'AWQI_Score']
# Class_output= ["CCME_WQI", "WAWQI_Quality", "IndWQI", "IWQI", "DWQI", "AWQI", "RWQI"]
```

```
# Separate input and output variables
X = data[input_features]
y_reg = data[Reg_output]
# y_class = data[Class_output]
```

```
y_reg = y_reg['IndWQI_Score']
```

```
y_reg
```

	IndWQI_Score
0	42.301154
1	78.931487
2	78.214201
3	78.192201
4	78.295059
...	...
29154	70.137143
29155	70.672857
29156	63.861429
29157	59.959011
29158	59.304121

29159 rows × 1 columns

dtype: float64

```
X.head()
```

	Alkalinity	Ammonia	BOD	Chloride	Conductivity	DO	orthophosphate	pH	Temp	Hardness	Color
0	314.0	0.033	1.2	27.3	711.0	52.50	0.019	7.80	10.4	370.0	24.0
1	14.0	0.033	1.2	10.0	71.0	61.85	0.019	7.42	17.8	13.4	35.0
2	17.0	0.033	1.2	11.5	79.0	61.80	0.019	7.67	18.1	15.8	29.0
3	18.0	0.033	1.2	11.3	78.0	62.40	0.019	7.63	17.8	15.9	31.0
4	19.0	0.033	1.2	10.5	76.0	66.05	0.019	7.33	20.1	15.4	23.0

```
X.describe()
```

	Alkalinity	Ammonia	BOD	Chloride	Conductivity	DO	orthophosphate	pH	
count	29159.000000	29159.000000	29159.000000	29159.000000	29159.000000	29159.000000	29159.000000	29159.000000	291
mean	139.858348	0.063573	1.276861	20.257858	363.039081	58.286184	0.068789	7.552057	
std	95.554543	0.441630	0.559890	19.127637	182.190019	26.946092	1.010133	0.581265	
min	0.000000	0.000000	0.000000	0.000000	33.000000	0.000000	-0.004000	4.700000	
25%	55.000000	0.033000	1.200000	15.600000	229.000000	49.400000	0.017000	7.200000	
50%	127.000000	0.033000	1.200000	18.600000	356.000000	54.900000	0.019000	7.700000	
75%	214.000000	0.033000	1.200000	22.000000	501.000000	75.000000	0.022000	8.000000	
max	442.000000	40.000000	16.000000	1260.000000	4200.000000	198.000000	70.000000	9.800000	6

```
y_reg.head()
```

	IndWQI_Score
0	42.301154
1	78.931487
2	78.214201
3	78.192201
4	78.295059

dtype: float64

```
y_reg.describe()
```

	IndWQI_Score
count	29159.000000
mean	57.031176
std	10.071092
min	27.006154
25%	48.702234
50%	55.200916
75%	65.287967
max	84.193418

dtype: float64

```
# Split the data into training and testing sets
X_train_reg, X_test_reg, y_train_reg, y_test_reg= train_test_split(X, y_reg, test_size=0.2, random_state=42)
```

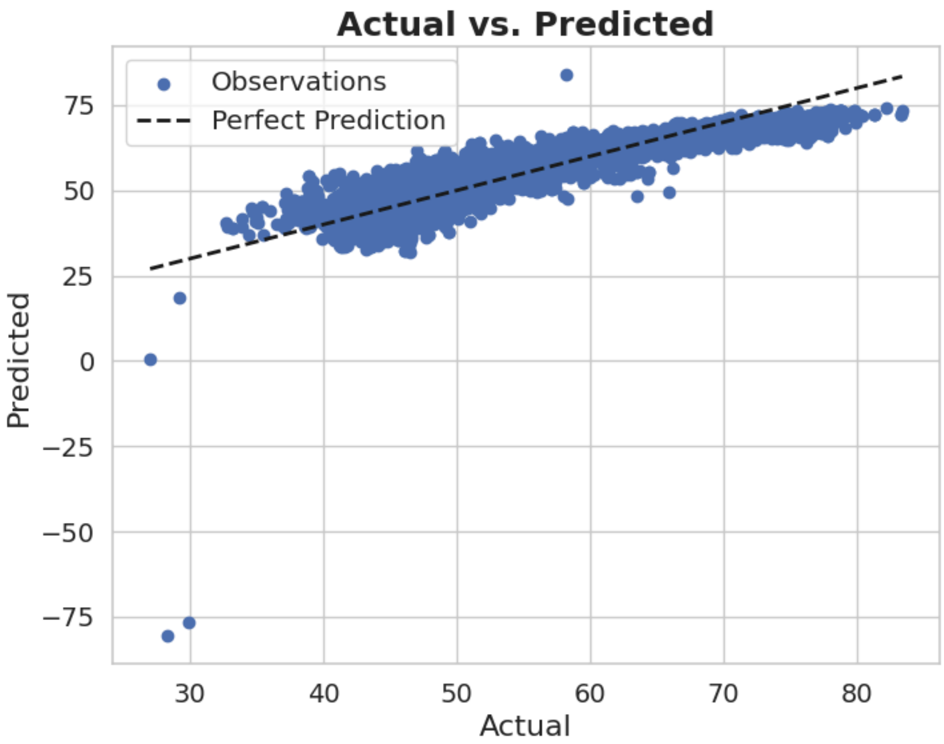
```
# Standardize the input features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_reg)
X_test_scaled = scaler.transform(X_test_reg)
```

1. Ridge Regression

```
from sklearn.linear_model import Ridge
from sklearn.metrics import (
    mean_absolute_error, mean_squared_error,
    mean_absolute_percentage_error, r2_score
)
import numpy as np

# Define the model with default parameters
ridge = Ridge()

# Fit the model
ridge.fit(X_train_reg, y_train_reg)

# Predict
y_pred = ridge.predict(X_test_reg)

# Evaluation
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Print results
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
MAE: 3.1926769762840457
MSE: 20.02633181088526
RMSE: 4.475078972586434
MAPE: 0.05945295437652003
R2 Score: 0.805295082297328
```

```python
import matplotlib.pyplot as plt

# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```



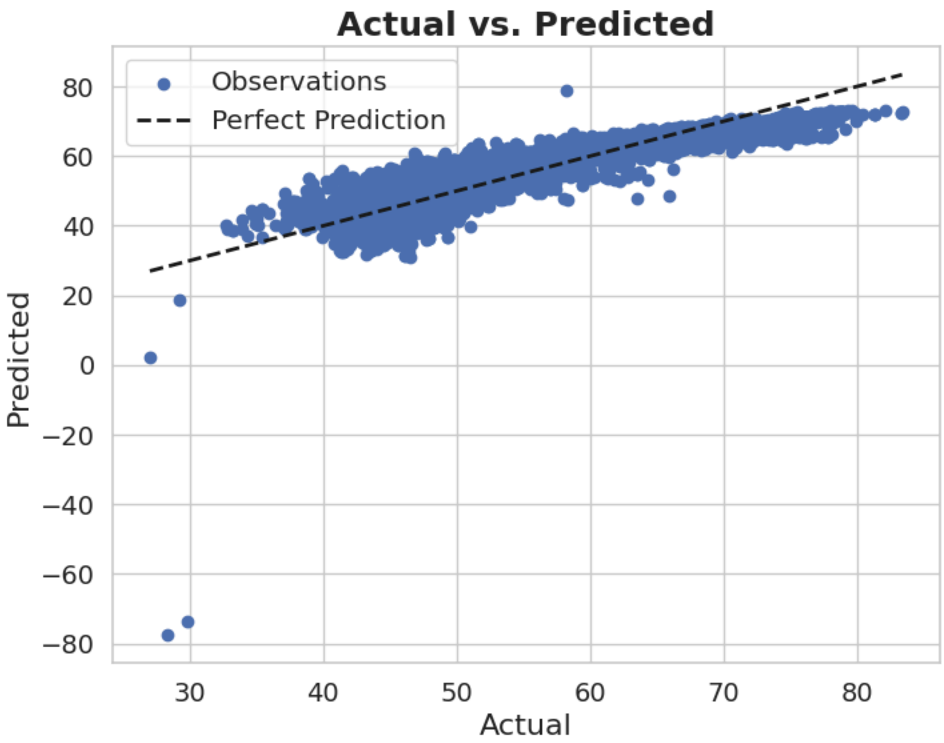
2. Lasso Regression

```python
from sklearn.linear_model import Lasso
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define the Lasso regression model with default parameters
lasso = Lasso(max_iter=10000)

# Fit the model
lasso.fit(X_train_reg, y_train_reg)

# Predict on test set
y_pred = lasso.predict(X_test_reg)

# Evaluation metrics
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Print results
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
MAE: 3.303857123372663
MSE: 20.952315798723824
```

```
RMSE: 4.577369965244651
MAPE: 0.0614563783079185
R2 Score: 0.7962922535292507
```

```
# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```

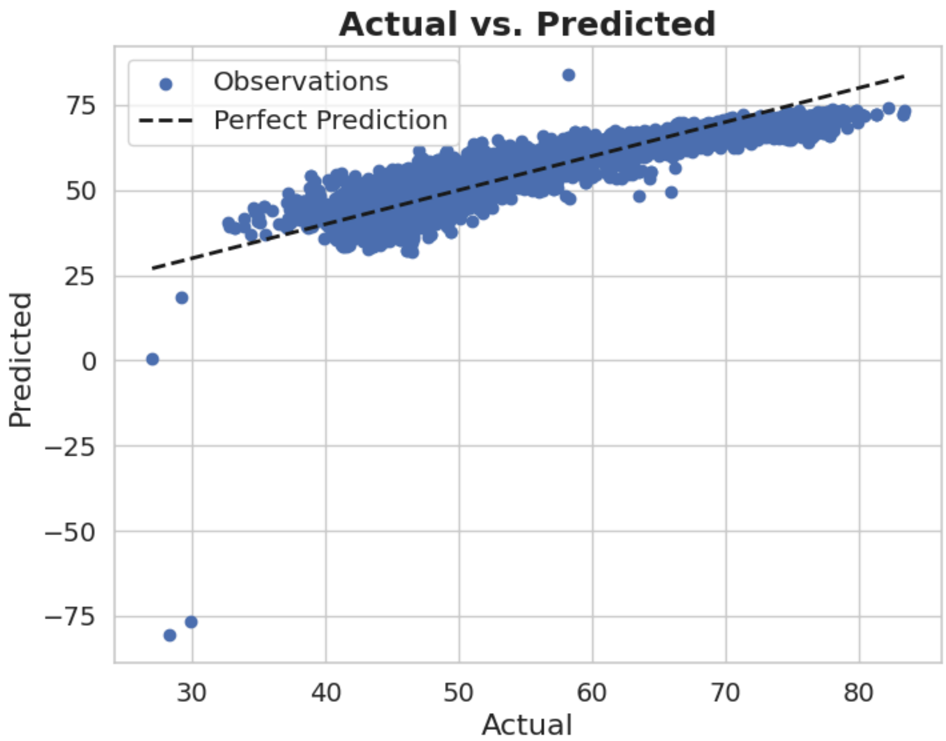


3. Linear Regression

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define model with default parameters
lr = LinearRegression()

# Fit model
lr.fit(X_train_reg, y_train_reg)

# Predict
y_pred = lr.predict(X_test_reg)

# Metrics
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Output
print("Model Parameters:", lr.get_params())
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
Model Parameters: {'copy_X': True, 'fit_intercept': True, 'n_jobs': None, 'positive': False}
MAE: 3.1926715824150103
MSE: 20.026280528287103
RMSE: 4.475073242784648
MAPE: 0.059452858236740846
R2 Score: 0.8052955808895889
```

```
# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```



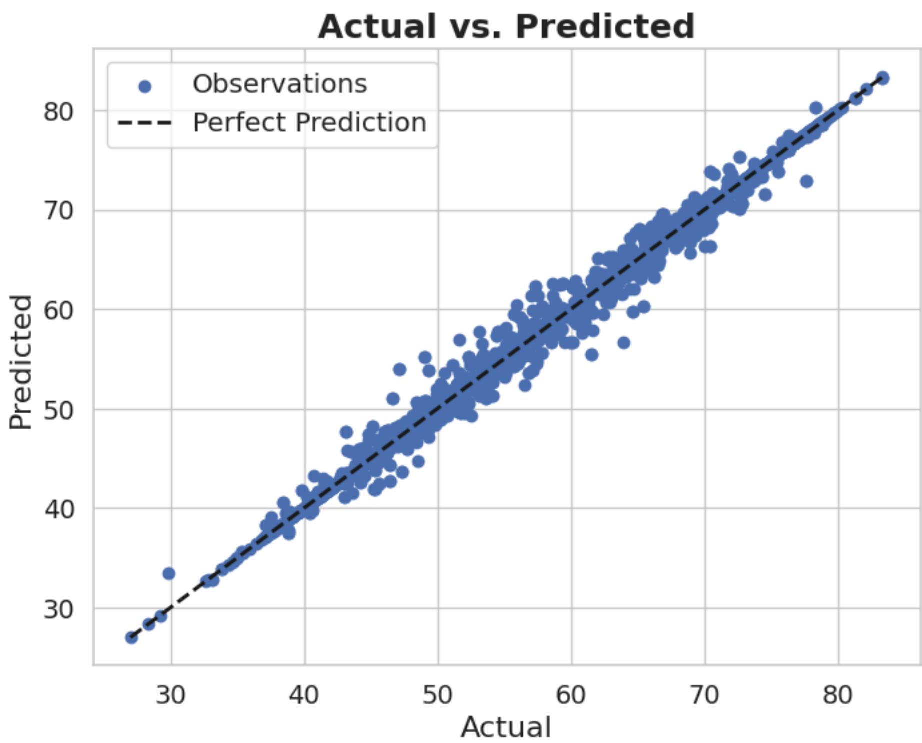
4. Decision Tree Regressor

```
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define the model with default parameters
dt = DecisionTreeRegressor(random_state=42)

# Fit model
dt.fit(X_train_reg, y_train_reg)

# Predict
y_pred = dt.predict(X_test_reg)

# Evaluation
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Output
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
```

```
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
MAE: 0.23014007290769586
MSE: 0.48169249925772184
RMSE: 0.6940407043234005
MAPE: 0.004125092290978586
R2 Score: 0.9953167709737542
```

```
# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```



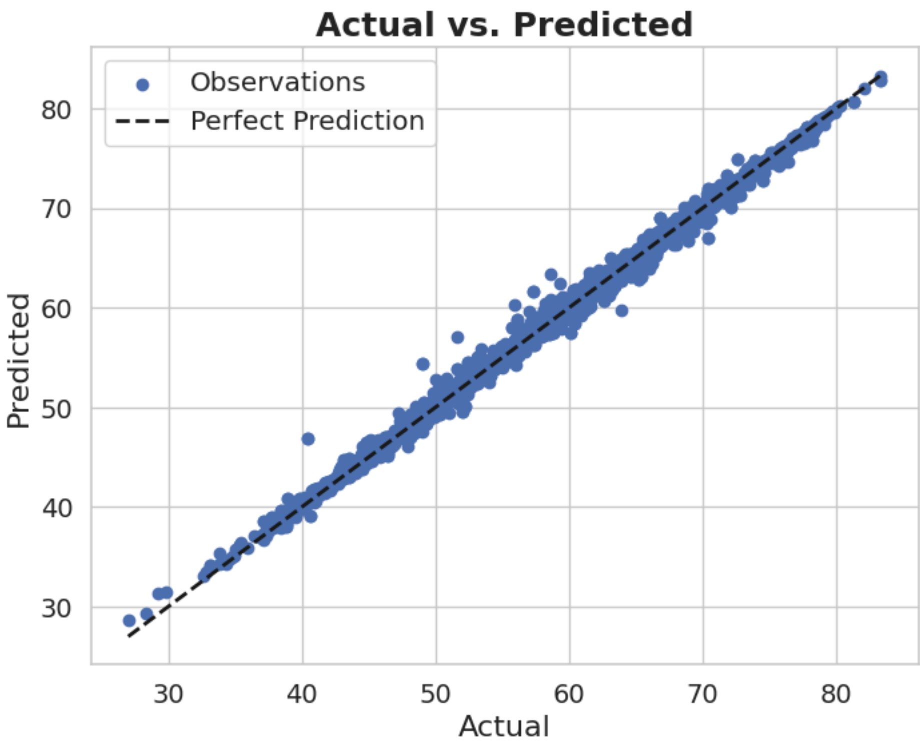
5. Random Forest Regressor

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define the model with default parameters
rf = RandomForestRegressor(random_state=42)

# Fit the model
rf.fit(X_train_reg, y_train_reg)

# Predict
y_pred = rf.predict(X_test_reg)

# Evaluation
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Results
print(f"MAE: {mae}")
print(f"MSE: {mse}")
```

```
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
MAE: 0.2822407728355598
MSE: 0.24593249399118725
RMSE: 0.49591581341109425
MAPE: 0.005067587709267185
R2 Score: 0.9976089347537456
```

```
# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```



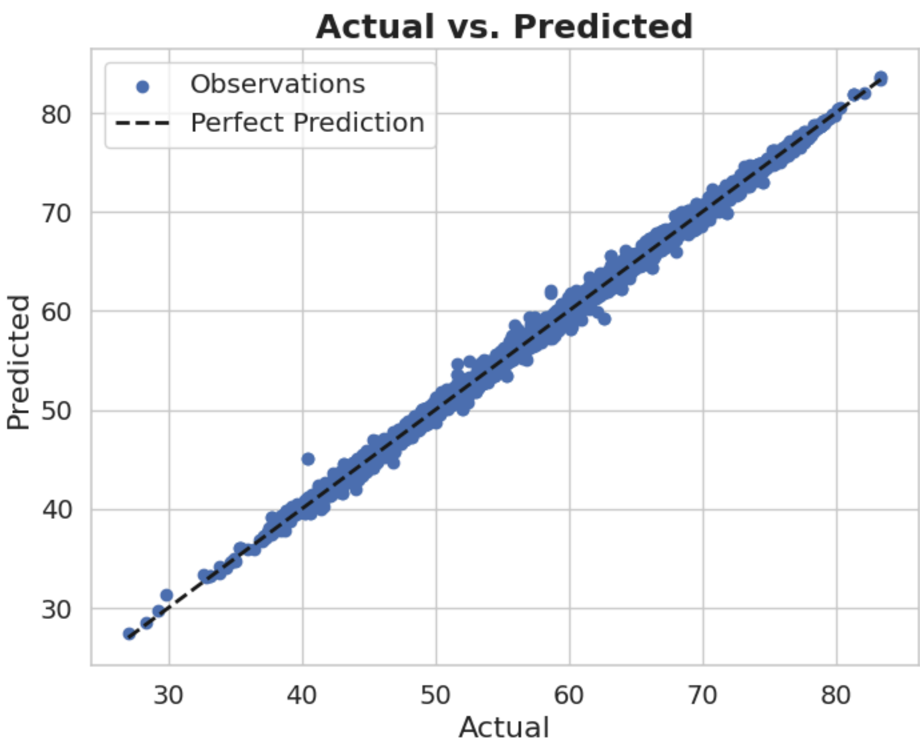
6. XGB Regressor

```
from xgboost import XGBRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define and train the model with default parameters
xgb_model = XGBRegressor(objective='reg:squarederror', random_state=42)
xgb_model.fit(X_train_reg, y_train_reg)

# Predict
y_pred = xgb_model.predict(X_test_reg)

# Evaluation
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Print results
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
```

```
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
MAE: 0.33736667137148546
MSE: 0.22140944291232398
RMSE: 0.4705416484354217
MAPE: 0.006076460733051686
R2 Score: 0.9978473587790347
```

```
# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```



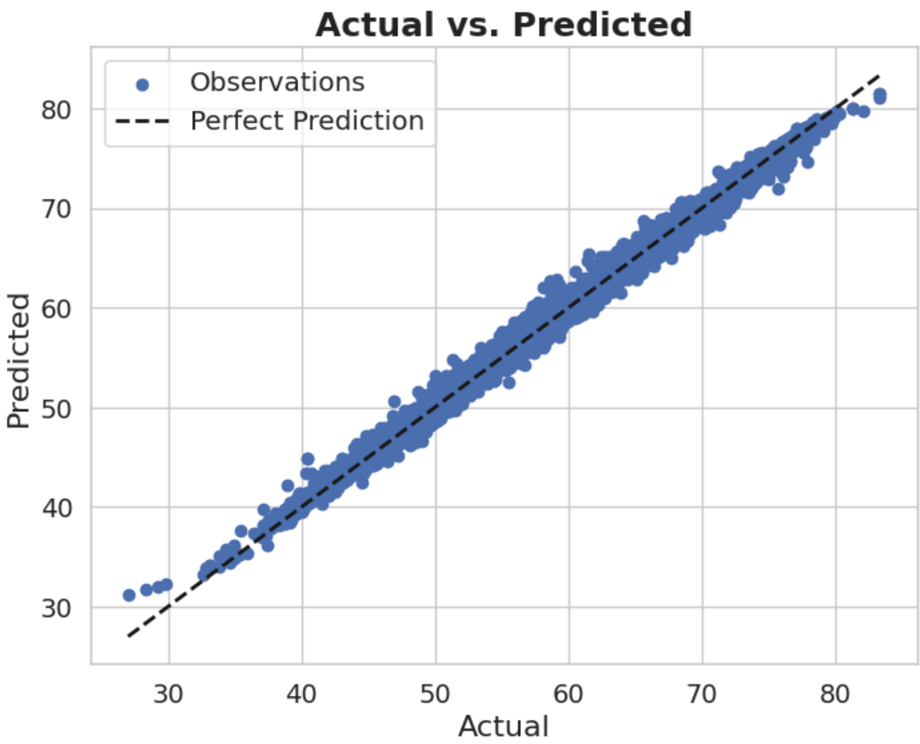
7. Gradient Boosting Regressor

```
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define the model with default parameters
gbr = GradientBoostingRegressor(random_state=42)

# Fit the model
gbr.fit(X_train_reg, y_train_reg)

# Predict
y_pred = gbr.predict(X_test_reg)

# Evaluation
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Output
print(f"MAE: {mae}")
print(f"MSE: {mse}")
```

```
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
MAE: 0.58597442466927
MSE: 0.6098616922256972
RMSE: 0.7809364200917366
MAPE: 0.010480890723776382
R2 Score: 0.994070652992463
```

```
# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```




```
# This is formatted as code
```

8. LightGBM Regressor

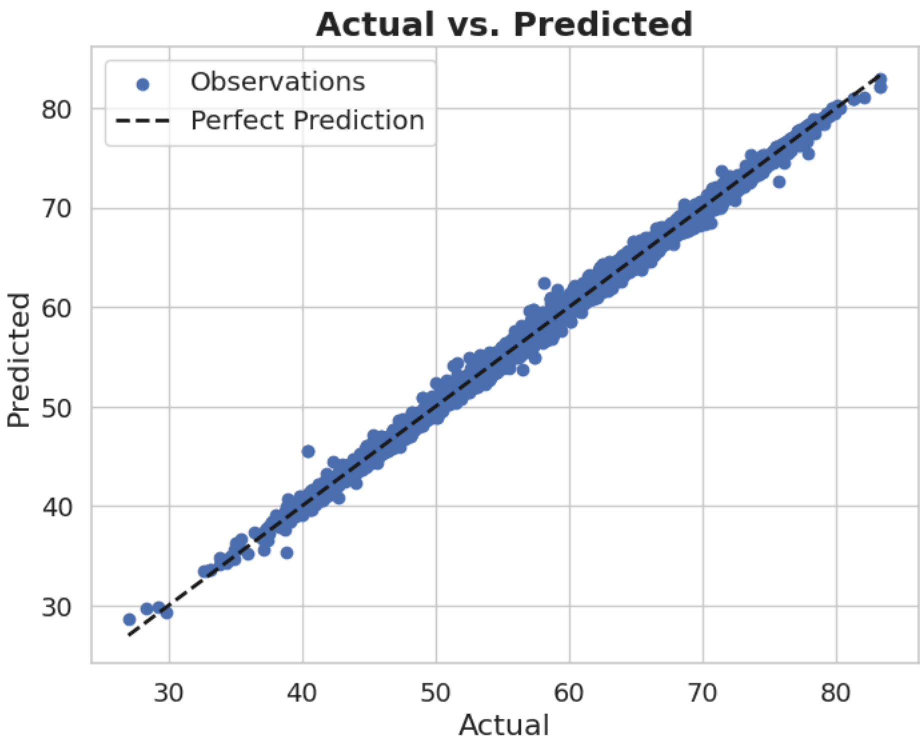
Double-click (or enter) to edit

```
import lightgbm as lgb
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define LightGBM Regressor with default parameters
lgb_reg = lgb.LGBMRegressor(random_state=42)

# Fit model on training data
lgb_reg.fit(X_train_reg, y_train_reg)

# Predict on test data
y_pred = lgb_reg.predict(X_test_reg)

# Evaluation metrics
mae = mean_absolute_error(y_test_reg, y_pred)
```

```
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

print("Best Parameters: Default parameters")
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of testing was 0.003060 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 2497
[LightGBM] [Info] Number of data points in the train set: 23327, number of used features: 11
[LightGBM] [Info] Start training from score 57.050431
Best Parameters: Default parameters
MAE: 0.3548290064925262
MSE: 0.24768401012851482
RMSE: 0.49767862132958335
MAPE: 0.006370030882513354
R2 Score: 0.9975919057337237
```

```
# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```



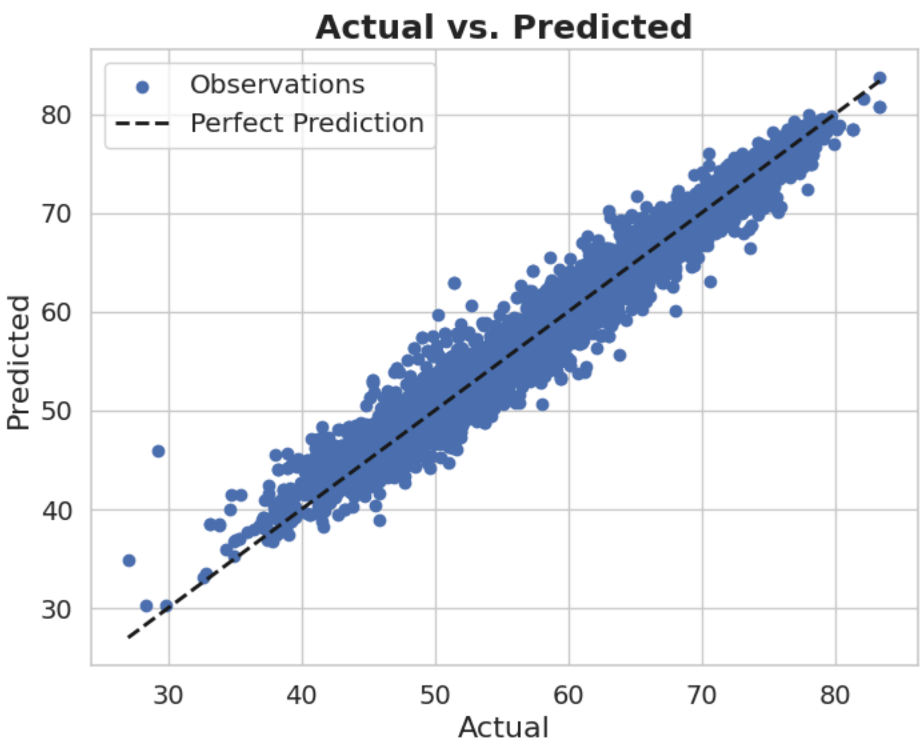
10 KNeighbors Regressor

```
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define the model with default parameters
knn = KNeighborsRegressor()

# Fit the model on training data
```

```
knn.fit(X_train_reg, y_train_reg)

# Predict on test set
y_pred = knn.predict(X_test_reg)

# Calculate metrics
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Print results
print("Using default KNeighborsRegressor parameters:")
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
Using default KNeighborsRegressor parameters:
MAE: 1.4469233515692788
MSE: 3.784677012329699
RMSE: 1.9454246354792823
MAPE: 0.026690410341473875
R2 Score: 0.9632036843703793
```

```
# Plotting the first output variable (CCME)
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```



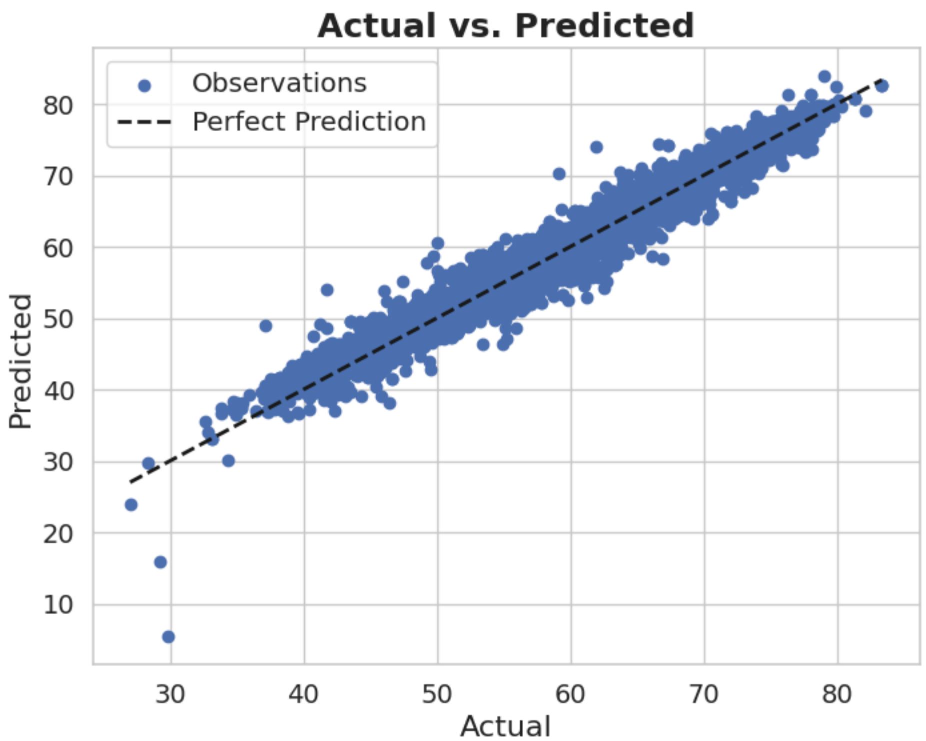
12. MLP Regressor

```
from sklearn.neural_network import MLPRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np

# Define the model with default parameters (except max_iter and random_state)
```

```
mlp = MLPRegressor(max_iter=1000, random_state=42)

# Fit model
mlp.fit(X_train_reg, y_train_reg)

# Predict
y_pred = mlp.predict(X_test_reg)

# Evaluation
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Print results
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
MAE: 1.4420920296963764
MSE: 3.7199261333780766
RMSE: 1.9287110030738344
MAPE: 0.026174858487036885
R2 Score: 0.9638332212559411
```

```
# Plotting the first output variable
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```



11. Support Vector Regression (SVR)

```
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error, mean_absolute_percentage_error, r2_score
import numpy as np
```

```
# Define SVR model with default parameters
svr = SVR()

# Fit the model on training data
svr.fit(X_train_reg, y_train_reg)

# Predict on test data
y_pred = svr.predict(X_test_reg)

# Evaluation metrics
mae = mean_absolute_error(y_test_reg, y_pred)
mse = mean_squared_error(y_test_reg, y_pred)
rmse = np.sqrt(mse)
mape = mean_absolute_percentage_error(y_test_reg, y_pred)
r2 = r2_score(y_test_reg, y_pred)

# Print results
print(f"MAE: {mae}")
print(f"MSE: {mse}")
print(f"RMSE: {rmse}")
print(f"MAPE: {mape}")
print(f"R2 Score: {r2}")
```

```
MAE: 1.7378055030698598
MSE: 5.954700039828338
RMSE: 2.440225407586016
MAPE: 0.03238325263818742
R2 Score: 0.9421057539569638
```

```
# Plotting the first output variable
plt.figure(figsize=(8, 6))
plt.scatter(y_test_reg, y_pred, color='b', label='Observations')
plt.plot([y_test_reg.min(), y_test_reg.max()],
         [y_test_reg.min(), y_test_reg.max()],
         'k--', lw=2, label='Perfect Prediction')
plt.xlabel('Actual', fontsize=16)
plt.ylabel('Predicted', fontsize=16)
plt.title('Actual vs. Predicted', fontsize=18, fontweight='bold')
plt.legend(fontsize='large')

plt.tick_params(axis='x', labelsize=14)
plt.tick_params(axis='y', labelsize=14)
plt.show()
```